

**1. Eigenbases**

$$\mathbf{L} = \mathbf{Q}_L \mathbf{\Lambda}_L \mathbf{Q}_L^\top$$
$$\mathbf{R} = \mathbf{Q}_R \mathbf{\Lambda}_R \mathbf{Q}_R^\top$$

**2. Rotate**

$$\tilde{\mathbf{G}} = \mathbf{Q}_L^\top \mathbf{G} \mathbf{Q}_R$$

**3. Adam**

per-coord adaptive  
in rotated basis

**4. Rotate back**

$$\mathbf{Q}_L \tilde{\mathbf{\Delta}} \mathbf{Q}_R^\top$$

*SOAP pipeline: Shampoo eigenbases + Adam adaptivity*